

Ankit Patil

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PROFESSIONAL SUMMARY

AI Engineer with 2+ years shipping governed agentic systems and multi-tenant retrieval platforms on AWS from vector search and knowledge graph pipelines to responsible AI guardrails with cryptographic audit trails. Built and deployed human-in-the-loop LLM orchestration serving live financial workflows, with hands-on depth in evaluation frameworks (RAGAS), inference cost optimization, and production observability.

Core Stack: Python · LangGraph · FastAPI · pgvector · Neo4j · AWS ECS Fargate

WORK EXPERIENCE

Invisia AI

New York, NY, USA

AI Engineer

Oct 2025 – Jun 2026

- Engineered a tenant-isolated vector search and ingestion pipeline using pgvector and Python, processing 50MB+ financial documents per tenant with strict namespace partitioning that eliminated cross-tenant data leakage in retrieval results
- Accelerated investor reporting cycles by 40%, as measured by compliance team turnaround benchmarks, by orchestrating a human-in-the-loop agentic workflow using LangGraph and GPT-4o with structured token budgeting to control per-report inference cost
- Built ToneGuard, a responsible AI guardrail enforcing citation-grounded faithfulness checks and content suppression, logging cryptographically immutable audit trails that cut compliance review time by replacing manual spot-checks with automated evaluation
- Deployed governed agentic microservices behind RBAC-enforced APIs on AWS ECS Fargate with Docker and GitHub Actions CI/CD, integrating a custom MCP tool server to enforce type-safe, policy-scoped agent tool execution in production

G5InfoTech

Remote, USA

AI Intern

Jan 2025 – Oct 2025

- Built a multi-turn conversational AI system using LangGraph with hybrid dense/sparse retrieval & cross-encoder reranking, benchmarked at 0.87 faithfulness via a RAGAS evaluation harness, automating 45% of internal support queries
- Reduced domain-specific hallucinations 3× by deploying locally-hosted Ollama models with structured prompting, persistent memory, and exception-aware retry logic — lowering inference costs versus cloud API calls while improving answer grounding
- Constructed entity-extraction & knowledge graph pipelines over internal documents using Neo4j & LLM-guided traversal, improving multi-hop answer accuracy by 35% vs flat vector baselines as measured by offline relevance evaluations

SafeSpace

Remote, USA

Software Engineering Intern

Aug 2024 – Dec 2024

- Shipped backend features and automated test suites using FastAPI and pytest under iterative code-review cycles, reducing post-deployment bugs by 20% as tracked across sprint retrospectives
- Architected REST and server-sent event streaming APIs integrated with a Next.js frontend to deliver real-time system state updates, compressing data triage workflows from hours to minutes

EDUCATION

State University of New York, Binghamton

Binghamton, NY, USA

Master of Science, Computer Science

Jan 2023 – May 2025

Savitribai Phule Pune University

Pune, India

Bachelor of Engineering, Computer Engineering

Aug 2018 – May 2022

PROJECTS

Incident Zero - Multimodal Agentic Security Platform | FastAPI, Next.js, Mistral, MCP, OCR, SSE

- Built an end-to-end agentic pipeline that accepts images, PDFs, and text evidence, runs OCR extraction and LLM-driven attack-graph generation, and outputs structured root-cause analysis handling modalities no single-model API covers natively
- Implemented a typed MCP tool registry with strict JSON schema validation and LLM function-calling, enforcing predictable agent behavior across 8+ investigative tools with zero unstructured fallback

WanderGenie - Multi-Agent AI Travel Planner | LangGraph, AWS Bedrock, GPT-4o-mini, pgvector, Neo4j

- Orchestrated three specialized LangGraph agents (planner, researcher, packager) coordinating through a shared state graph to generate multi-step itineraries, with agent-level evaluation tracking via Langfuse
- Benchmarked hybrid retrieval (dense + BM25 + Neo4j graph traversal) offline against relevance and coherence metrics, selecting the chunking and reranking strategy that improved route accuracy by 40% over dense-only baselines

SKILLS

Languages & Databases: Python, SQL, TypeScript, FastAPI, React.js, PostgreSQL, MongoDB

AI & Agents: LangGraph, LangChain, LlamaIndex, Agentic Orchestration, MCP Servers, Human-in-the-Loop, Prompt Engineering, Structured Outputs

Retrieval & Evaluation: pgvector, Neo4j, FAISS, Hybrid Retrieval, Cross-Encoder Reranking, Graph RAG, RAGAS, Langfuse, LLMops

LLMs: OpenAI GPT-4o, Claude, Mistral, Ollama, Fine-tuning (LoRA/QLoRA via PEFT), HuggingFace Transformers

Cloud & Infra: AWS (ECS Fargate, Lambda, S3, SageMaker, Bedrock), Azure, Docker, Kubernetes, Terraform, GitHub Actions CI/CD

Certifications: NVIDIA Generative AI - Associate & Professional · NVIDIA Agentic AI - Professional